

LEADER & ACHIEVER COURSE

PHASE : MLI,J,K,M,N,O,R,S,M4L8A,MAZG,H,I,J,K,L,M,T,U,M1A5A,
M2A5B, M4AA2A, M2AP1A,M2AP1B

TARGET : PRE-MEDICAL 2020

Test Type : MAJOR

Test Pattern : NEET(UG)

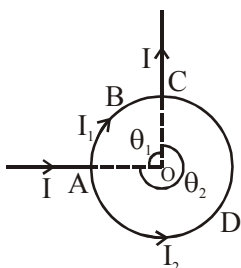
TEST DATE : 23 - 03 - 2020

TEST SYLLABUS : 05 & 06

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	1	1	4	4	4	4	1	4	4	1	2	3	2	1	4	4	2	4	2	1	1	4	4	3	3	4	1	4	3	3
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	4	1	4	2	3	4	4	4	1	2	4	2	1	3	1	3	1	4	4	1	4	3	4	2	1	2	4	3	1	1
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	4	2	1	1	1	4	3	3	2	4	1	4	3	2	2	3	2	4	4	1	4	1	1	4	3	1	1	2	3	3
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	2	3	2	1	4	4	1	4	2	4	2	4	4	3	2	2	2	2	3	3	3	1	2	4	3	2	3	3	3	4
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	2	2	4	1	4	1	2	3	4	4	1	3	4	4	1	2	3	3	3	3	2	2	1	3	3	1	2	2	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	4	3	1	2	4	4	4	1	3	3	2	4	3	4	1	3	2	4	2	2	2	4	1	1	2	1	3	3	2	2

HINT – SHEET

1. Let l_1, l_2 be the lengths of the two parts ABC and ADC of the conductor and ρ be resistance per unit length of the conductor.



The resistance of the part ABC will be : $R_1 = \rho l_1$
The resistance of the part ADC will be : $R_2 = \rho l_2$
Potential difference across

$$AC = I_1 R_1 = I_2 R_2$$

$$I_1 \rho l_1 = I_2 \rho l_2$$

$$\text{or } I_1 l_1 = I_2 l_2 \quad \dots(i)$$

Magnetic field at O due to a current I_1 flowing in ABC is given by :

$$B_1 = \frac{\mu_0 I_1 \theta_1}{4\pi R} = \frac{\mu_0 I_2 l_2}{4\pi R^2} \odot \quad (\text{As } l_1 = \theta_1 R)$$

Magnetic field at O due to current I_2 flowing in ADC is given by :-

$$B_2 = \frac{\mu_0 I_2 \theta_2}{4\pi R} = \frac{\mu_0 I_2 l_1}{4\pi R^2} \odot \quad (\text{As } l_2 = \theta_2 R)$$

$$\text{or } B_2 = \frac{\mu_0 I_1 l_1}{4\pi R^2} \odot \quad [\text{Using eqn.(i)}]$$

The resultant magnetic field induction at O is,

$$B = B_2 - B_1 = 0 \quad (\text{As } B_2 = B_1)$$

$$2. \quad E = \frac{\Delta\phi}{\Delta t} \Rightarrow \Delta t = \frac{\Delta\phi}{E} = \frac{50 \times 0.02 \times 100 \times 10^{-4}}{0.1} = 0.1 \text{ s}$$

$$3. \quad \therefore V_L = V_C \Rightarrow \text{Resonance condition}$$

$$\therefore V_R = V = 220 \text{ V}$$

$$4. \quad AP = A_V \beta \Rightarrow \frac{A_P}{A_V} = \beta$$

6. Magnetic field on axis of a circular current carrying coil is

$$B_x = \frac{\mu_0 I a^2}{2(a^2 + x^2)^{3/2}}$$

$$B_x = \frac{\mu_0}{4\pi} \frac{2\pi I a^2}{(x^2 + a^2)^{3/2}}$$

7. $q = \frac{\Delta\phi}{R} = \frac{60-10}{100} = 0.5 \text{ C}$

8. $P_{\text{watt less}} = \frac{VI}{2} \sin \phi \quad \left\{ \phi = \frac{\pi}{6} \right\}$

$$= \frac{100 \times 100}{2} \times \frac{1}{2} = 2500 \text{ W}$$

9. 108 electrons enter the emitter in 10^{-8} s

i.e., $I_E = \frac{10^8 \times 1.6 \times 10^{-19}}{10^{-8}} \text{ A}$
 $= 172.8 \times 10^{-11} \text{ A}$

$\therefore 1\% \text{ of } I_E \text{ is lost in base, i.e., } I_B = \frac{I_E}{100}$

$\Rightarrow \frac{99}{100} I_E \text{ enters the collector}$

$\Rightarrow I_C = 0.99 I_E$

Current amplification factor

$$\beta = \frac{I_C}{I_B} = \frac{0.99 I_E}{0.01 I_E} = 99$$

10. Number of photons emitted per second

$$n = \frac{P}{h\nu} = \frac{10 \times 10^3}{6.6 \times 10^{-34} \times 880 \times 10^3} = 1.72 \times 10^{31}$$

11. Susceptibility is negative

12. As magnetic field is steady and not changing with time so no EMF is induced in the ring in any of the two states mentioned.

13. In circuit (b), each p-n junction is forward biased, same current will flow giving equal potential difference across each p-n junction diode. In circuit (c), each p-n junction is reverse biased, hence same current flows, giving same potential difference across p-n junctions.

14. NOT gate at output of OR gate this combination is NOR gate

15. No light after reflection from mirror will reach person's eye.

16. $R = \frac{mV}{qB} \Rightarrow R = \frac{P}{qB}$

$$R \propto \frac{1}{q}$$

18. Current carried by the holes, $I_h = \frac{1}{5} I$

when I is the total current

Current carried by the electrons, $I_e = \frac{4}{5} I$

$$v_e = 2v_h$$

where v_e and v_h are the drift speeds of electrons and holes respectively

$$I_e = n_e A e v_e \quad \dots (i)$$

$$I_h = n_h A e v_h \quad \dots (ii)$$

when n_e and n_h are the number densities of electrons and holes respectively

$$\therefore \frac{I_e}{I_h} = \frac{n_e A e v_e}{n_h A e v_h} \text{ or } \frac{n_e}{n_h} = \frac{I_e}{I_h} \times \frac{v_h}{v_e}$$

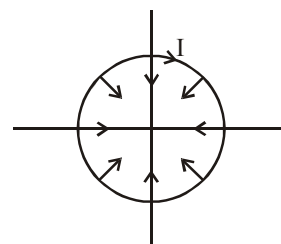
$$\frac{n_e}{n_h} = \frac{\frac{4}{5} I}{\left(\frac{1}{5} I\right)} \times \left(\frac{v_h}{2v_h}\right) = 2$$

19. x-intercept = $v_0 = 5 \times 10^{14} \text{ Hz}$

$$\therefore \lambda = \frac{c}{v_0} = \frac{3 \times 10^8}{5 \times 10^{14}} = 6 \times 10^{-7} \text{ m} = 6000 \text{ \AA}$$

20. Focal length of spherical mirror depends only on its radius not on medium.

21. Force on small elements of loop will be as shown, by right hand palm rule so it will be contract



22. $\eta = \frac{P_o}{P_i} \times 100 = \frac{11 \times 90}{220 \times 5} \times 100 = 90\%$

23. For full wave rectifier $\eta = \frac{81.2}{1 + \frac{r_f}{R_L}}$

$$\Rightarrow \eta_{\text{max}} = 81.2 \% \quad (r_f \ll R_L)$$

24. $E = nh\nu \Rightarrow \nu \propto \frac{1}{n} \Rightarrow \frac{n_1}{n_2} = \frac{\gamma_2}{\gamma_1}$

25. mlt Refractive Index dsmedim esa tc light igqprh gsrksmukgh Angle of refraction is same as angle of incidence for parallel glass

26. As the point O is along the length of the straight wires, so the field at O due to them will be zero and hence.

(A) $\vec{B} = \frac{\mu_0}{4\pi} \pi I \left[\frac{1}{R_1} - \frac{1}{R_2} \right] \odot$ &

$\vec{M} = NI\vec{A} = 1 \times I \left[\frac{1}{2} \pi R_2^2 \otimes + \frac{1}{2} \pi R_1^2 \odot \right]$

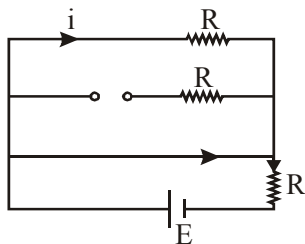
$= \frac{1}{2} \pi I [R_2^2 - R_1^2] \otimes$

(B) Following as in case (A), in this situation,

$\vec{B} = \frac{\mu_0}{4\pi} \pi I \left[\frac{1}{R_1} + \frac{1}{R_2} \right] \otimes$ and, $\vec{M} = \frac{1}{2} \pi I [R_2^2 + R_1^2] \otimes$

27. $\frac{E_0}{B_0} = C. = \frac{w}{k} \Rightarrow KE_0 = \omega B_0$

28. Diodes D_1 and D_3 are forward biased and D_2 is reverse biased so the circuit can be redrawn as follows due to short circuit current i will be zero.



29. $\lambda = \frac{h}{p} \Rightarrow \frac{\Delta\lambda}{\lambda} = \frac{\Delta p}{p}$

$\Rightarrow \frac{0.25}{100} = \frac{p_0}{p} \Rightarrow p = 400 p_0$

31. Work done = $MB (\cos\theta_1 - \cos\theta_2)$

$\theta_1 = 0$ and $\theta_2 = 180^\circ$

$W = MB (\cos 0 - \cos 180^\circ) = 2MB$

32. Electric flux = $\phi_E = E \cdot \pi r^2$

using ampere – maxwell equation

$\oint \vec{B} \cdot d\vec{l} = \mu_0 \epsilon_0 \frac{d\phi_E}{dt}$

$B(2\pi r) = \mu_0 \epsilon_0 \pi r^2 \frac{dB}{dt}$

$B = \frac{\mu_0 \epsilon_0 r}{2} \frac{dB}{dt} = \frac{1}{9 \times 10^{16} \times 2} \times 10^{10}$
 $= 5.56 \times 10^{-8} \text{ T}$

33. For figure (i) Time constant = $2RC$

By $q = CV e^{-t/2RC}$

$q = \frac{CV}{e^{1/2}}$

For figure (ii) diode is in reverse bias current = 0

$q = CV$ (remain constant)

34. Rate of disintegration $\frac{dN}{dt} = 10^{17} \text{ s}^{-1}$

Half life $T_{1/2} = 1445 \text{ year}$

$= 1445 \times 365 \times 24 \times 60 \times 60 = 4.55 \times 10^{10} \text{ sec}$

Now decay constant

$\lambda = \frac{0.693}{T_{1/2}} = \frac{0.693}{4.55 \times 10^{10}} = 1.5 \times 10^{-11} \text{ per sec}$

The rate of disintegration

$\frac{dN}{dt} = \lambda \times N_0 \Rightarrow 10^{17} = 1.5 \times 10^{-11} \times N_0$

$\Rightarrow N_0 = 6.6 \times 10^{27}$

35. Position of first minima = position of third maxima,

i.e., $\frac{1 \times \lambda_1 D}{d} = \frac{(2 \times 3 + 1) \lambda_2 D}{d} = \lambda_1 = 3.5 \lambda_2$

36. Magnetic potential at a distance d from the bar magnet on its axial line

$V = \frac{\mu_0 M}{4\pi d^2} \Rightarrow V \propto M \frac{V_1}{V_2} = \frac{M_1}{M_2}$

$\frac{V}{V_2} = \frac{M}{M/4} \Rightarrow V_2 = V/4$

37. $V = 220 \sin(100 \pi t)$

Time taken to reach from
max. to r.m.s. value.

$$= \frac{T}{8}$$

$$= \frac{1}{400} = 2.5 \times 10^{-3} \text{ s}$$

38. Voltage across R_2 is less than breakdown voltage (V_z) so no breakdown occurs & $i_z = 0$

39. From COLM - $m_1 \times 8 = m_2 \times 1$ ($m_1 v_1 = m_2 v_2$)

$$\frac{m_1}{m_2} = \frac{1}{8}$$

$$R = R_0(A)^{1/3}$$

$$\frac{R_1}{R_2} = \frac{R_0(A_1)^{1/3}}{R_0(A_2)^{1/3}} = \frac{1}{2}$$

40. According to Malus law, $I = I_0 \cos^2 \theta$

$$\text{After 2nd Polaroid, } I = \frac{I_0}{2} \cos^2 60^\circ = \frac{I_0}{8}$$

$$\text{After 3rd Polaroid, } I = \frac{I_0}{8} \cos^2 30^\circ = \frac{3I_0}{32}$$

$$41. T = 2\pi \sqrt{\frac{I}{MB_H}}; \therefore \frac{T_1}{T_2} = \sqrt{\frac{(B_H)_2}{(B_H)_1}} \Rightarrow T_2 = T_1 \sqrt{\frac{(B_H)_1}{(B_H)_2}}$$

$$\text{Here } n_1 = 30 \text{ oscillations/min} = \frac{1}{2} \text{ oscillations/sec}$$

$$\therefore T_1 = \frac{1}{n_1} = 2 \text{ sec}$$

$$\therefore T_2 = 2 \sqrt{\frac{B_H}{2B_H}} = 2 \times \frac{1}{\sqrt{2}} = \sqrt{2} \text{ sec}$$

42. $Z = 5 \Omega \quad R = 4 \Omega$

$$X_L = \sqrt{Z^2 - R^2} = 3 \Omega$$

43. $i = \frac{20V - 15V}{10k\Omega} = 0.5 \text{ mA}$

44. $T_n = T$

$$A = A_0 \left(\frac{1}{2} \right)^{\frac{t}{T}}$$

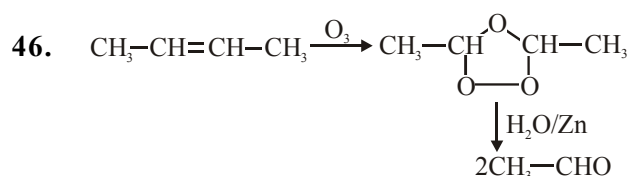
$$A_0 = A 2^{\frac{t}{T}}$$

45. $x_n = 2n \left(\frac{D\lambda}{2d} \right)$

$$\text{or } \frac{x_n}{D} = \frac{n\lambda}{d}$$

$$\therefore \sin \theta = \frac{n\lambda}{d} = \frac{3 \times 589 \times 10^{-9}}{0.589}$$

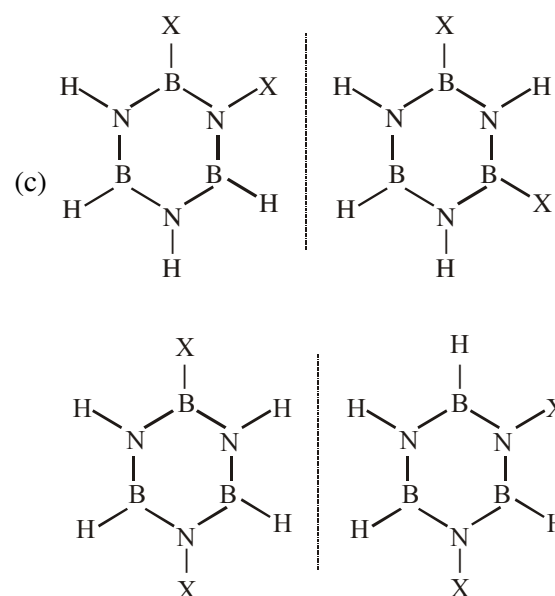
$$\text{or } \theta = \sin^{-1}(3 \times 10^{-6})$$



48. AgNO_3 react with anilinium hydrogen chloride.

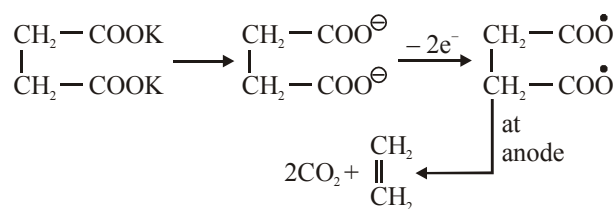
49. (a) fact

(b) Borax bead for transition element

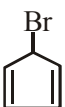


50. complementary color

51. Kolbe's electrolytic synthesis



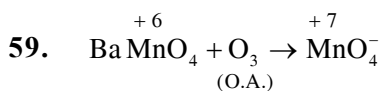
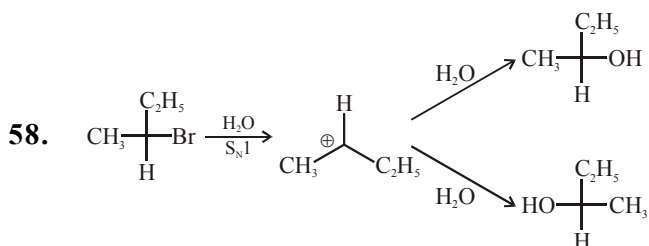
52. Fact

53.  not give S_N1 reaction.

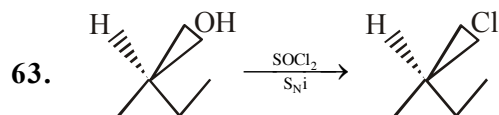
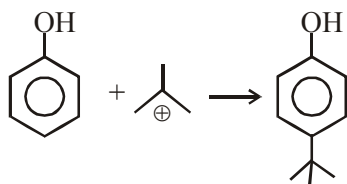
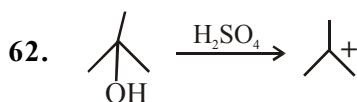
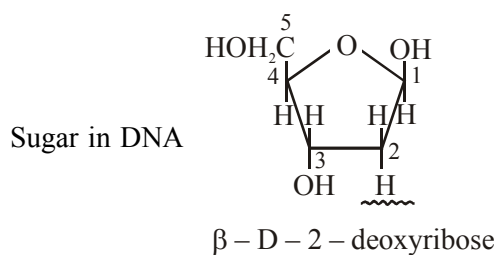
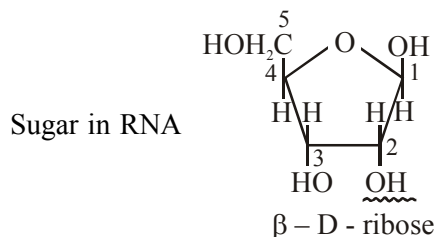
54. Ostwald process

55. $[Pt(NH_3)_2Cl_4] \Rightarrow Pt^{+4} = 6 \text{ C.N.}$

57. IPSO substitution

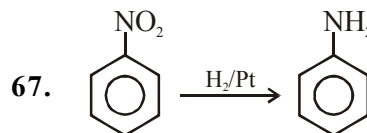


61. NCERT XIIth, Page No. # 427



64. $Cr^{+3} \Rightarrow 4s^0 3d^3 \Rightarrow t_{2g}^3$ half filled more stable

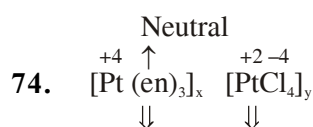
66. Starch is a polymer of α - D - Glucose



69. $V(CO)_6$ $[V(CO)_6]^-$
 extent of synergic bonding more
 $\therefore$ C-O B.L. $\propto$ -ve charge

71. Fact

72. TNT is 2,4,6-Trinitrotoluene



C.N. = 6 C.N = 4

O.S. = +4 O.S. = +2

$[Pt(en)_3]_2$ $[PtCl_4]_4$

$\Rightarrow x + y$

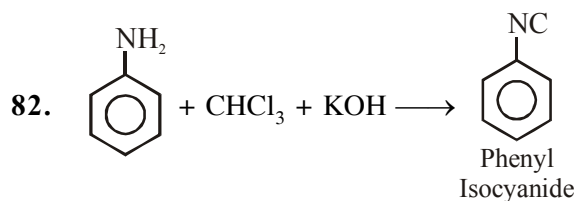
$\Rightarrow 2 + 4$ or $1 + 2 = 3$

76. RAYON is a semi-synthetic polymer

77. Hoffmann bromamide reaction

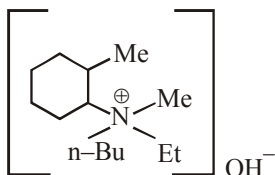
79. $[Co(NH_3)_6]^{+3}$ $[Co(ONO)_6]^{-3}$

81. Tegamet is a histamine receptor.



83. Pyrosilicate $\rightarrow$ basic unit $Si_2O_7^{-6}$

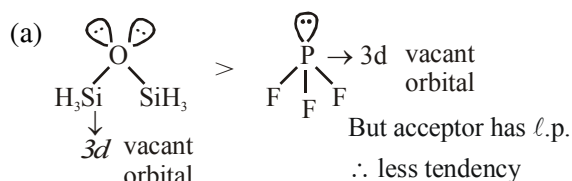
86.



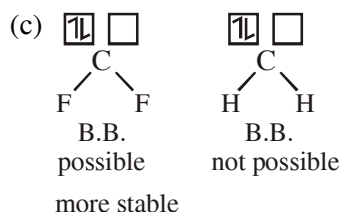
in above compounds ethyl group have most acidic β -H so formed $\text{CH}_2 = \text{CH}_2$

87. Only primary amine gie mustard oil reaction.

88.



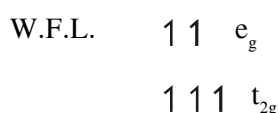
(b) $\text{NH}_3 > \text{N}(\text{SiH}_3)_3$ B.B. ∴ less tendency to donate



(d) H_3PO_4 Basicity $\Rightarrow$ 3
 $\therefore$ can form 3 type of salt.

89. High spin complex

↓



$$\text{CFSE} = 2\left(\frac{3}{5}\Delta_0\right) + 3\left(-\frac{2}{5}\Delta_0\right) + P \Rightarrow 0$$

$$\mu_s = \sqrt{n(n+2)} \text{ B.M.}$$

$$\Rightarrow \sqrt{5(5+2)} \text{ B.M.} \Rightarrow \sqrt{35} \Rightarrow 5.92 \text{ B.M.}$$

90. As the size increases bond length increases.

91. NCERT, Pg# 146 to 149

92. NCERT, Pg# 158

HS-6/7

94. NCERT-XII, Pg# 174

95. NCERT XII, Pg. # 204

99. NCERT, Pg. # 259

101. NCERT, Pg# 147

102. NCERT, Pg# 159

103. NCERT-XII, Pg# 96

104. NCERT-XII, Pg# 176

105. NCERT XII, Pg. # 181

109. NCERT, Pg. # 260

111. NCERT, Pg# 149, 150

112. NCERT, Pg# 159

113. NCERT-XII, Pg# 99

114. NCERT XII, Pg. # 177

115. NCERT XII, Pg. # 182

117. NCERT, Pg. # 242

118. Rate of heat loss directly proportional to surface volume ratio.

121. Module-1, Pg# 176

122. Module-1, Pg# 194

123. NCERT-XII, Pg# 104

124. NCERT XII, Pg. # 176,175,177

125. NCERT XII, Pg. # 185

129. NCERT, Pg. # 264

131. NCERT, Pg# 151

132. NCERT-XII, Pg# 70

Green pod and violet flower

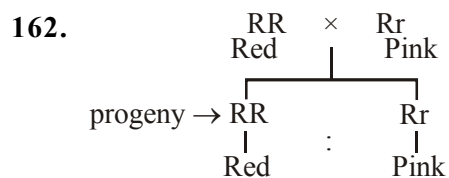
133. NCERT-XII, Pg# 111

134. NCERT XII, Pg. # 194

135. NCERT XII, Pg. # 208

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139. NCERT, Pg. # 265
141. NCERT, Pg# 152
142. $2^3 = 8$
143. NCERT-XII, Pg# 117
144. NCERT XII, Pg. # 195
145. NCERT XII, Pg. # 209
149. NCERT, Pg. # 260, Fig. 15.1
150. NCERT, Pg. # 275, 276, 277
151. NCERT, Pg# 150, 154
153. NCERT-XII, Pg# 122
154. NCERT XII, Pg. # 198
155. NCERT XII, Pg. # 211
157. NCERT, Pg. # 246
161. Module-1, Pg# 186, 187



163. NCERT-XII, Pg# 120
164. NCERT XII, Pg. # 201
165. NCERT XII, Pg. # 213
170. NCERT, Pg. # 229
171. Module-1, Pg# 179
172. NCERT-XII, Pg# 81
173. NCERT-XII, Pg# 171
174. $2^3 = 8$ molecules
175. NCERT XII, Pg. # 214
178. NCERT, Pg. # 253, Fig. 14.6
179. NCERT, Pg. # 282
180. NCERT, Pg. # 229, Fig. 13.5